

NUMPY: STUDENT SUBJECT AVERAGES

AIM

To use a NumPy array to calculate the average marks in Math, Science, English, and History for four students, and identify the subject with the highest average.

ALGORITHM

1. Start.
2. Import the NumPy library.
3. Create a 4×4 NumPy array containing the marks of four students. Each row represents a student, and the columns are ordered as Math, Science, English, and History.
4. Calculate the average of each column using `numpy.mean()` with `axis=0`.
5. Use `numpy.argmax()` to find the index of the highest average.
6. Use the index to display the subject name and its average.
7. Stop.

PROGRAM (PYTHON)

```
import numpy as np

# Columns: Math, Science, English, History
student_scores = np.array([
    [80, 75, 85, 70],
    [90, 80, 78, 88],
    [70, 85, 90, 75],
    [85, 90, 82, 80]
])

subjects = ["Math", "Science", "English", "History"]

# Calculate average marks for each subject (column)
averages = np.mean(student_scores, axis=0)

# Find the subject with the highest average
highest_index = np.argmax(averages)

print("Average marks for each subject:")
for i in range(len(subjects)):
    print(f"{subjects[i]}: {averages[i]:.2f}")

print("\nSubject with the highest average:",
      subjects[highest_index])
print("Highest average:", f"{averages[highest_index]:.2f}")
```

INPUT

The 4×4 student_scores array:

Math	Science	English	History
80	75	85	70
90	80	78	88
70	85	90	75
85	90	82	80

OUTPUT

```

# Average marks for each subject
Average marks for each subject:
Math: 81.25
Science: 82.50
English: 83.75
History: 78.25

# Subject with the highest average
Subject with the highest average: English
Highest average: 83.75

```

RESULT

The average marks for all four subjects were calculated successfully using NumPy. English has the highest average score of 83.75.